

# Assignment 1

## Question 1: Decision Tree Construction (AND Function) [CLO- 1]

Construct a decision tree for a 3-variable AND function with inputs:

$$(x_1, x_2, x_3) \in \{0,1\}^3$$

and output:

$$y = x_1 \wedge x_2 \wedge x_3$$

Tasks:

1. Create the full dataset (all 8 combinations).
2. Using Information Gain, determine:
  - The best feature for the root node.
3. Draw the final tree clearly.

## Question 2: Gradient Descent on a Quadratic Loss [CLO- 1]

Consider the following loss function:

$$L(w) = (w - 3)^2 + 2$$

You are given:

- Initial value:  $w_0 = 0$
- Learning rate:  $\eta = 0.1$

Tasks:

1. Compute the gradient of the loss function.
2. Perform two iterations of gradient descent:
  - Compute  $w_1$
  - Compute  $w_2$
3. Compute the loss at each step:
  - $L(w_0), L(w_1), L(w_2)$

### Question 3: K-Means Clustering (Unsupervised Learning) [CLO-2]

You are given the following 2D dataset:

$$X = \{(1,1), (2,1), (4,3), (5,4)\}$$

Initial cluster centroids are:

$$\mu_1 = (1,1), \mu_2 = (5,4)$$

Tasks:

1. Assign each data point to the nearest centroid using Euclidean distance.
2. Compute the new centroids after reassignment.
3. Perform one more iteration (assignment + update).
4. Report the final cluster assignments.

### Question 4: Soft Margin SVM and Slack Variables [Ungraded]

In hard-margin SVM, the constraint is:

$$y_i(w^T x_i + b) \geq 1$$

In soft-margin SVM, this is modified using slack variables:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

where  $\xi_i \geq 0$ .

Tasks:

1. Explain the role of slack variables  $\xi_i$ .
2. Describe how slack variables allow violations of the margin constraint.
3. Explain how this modification helps in handling:
  - Non-separable data
  - Noisy data
4. Discuss the role of parameter  $C$  in controlling these violations.